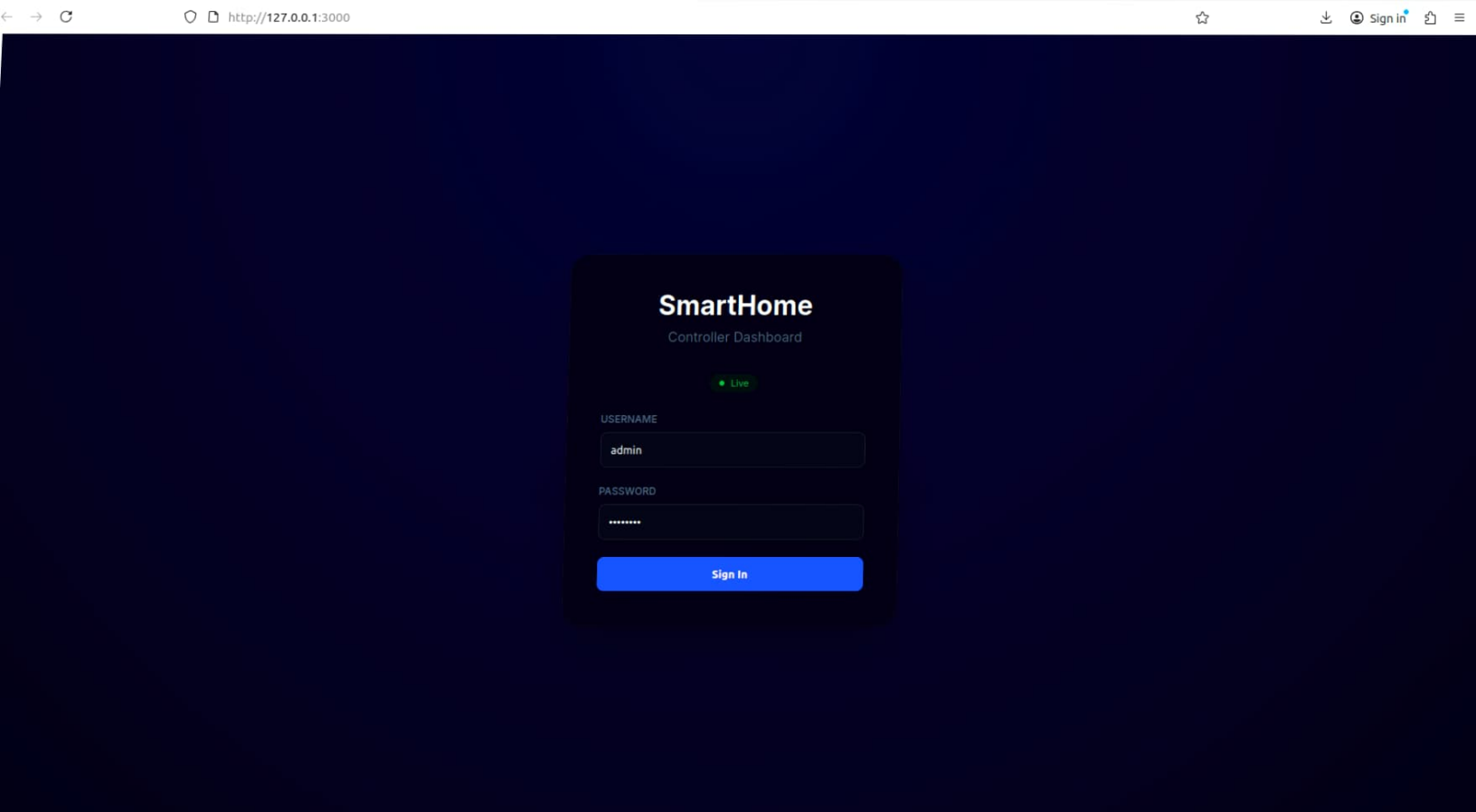
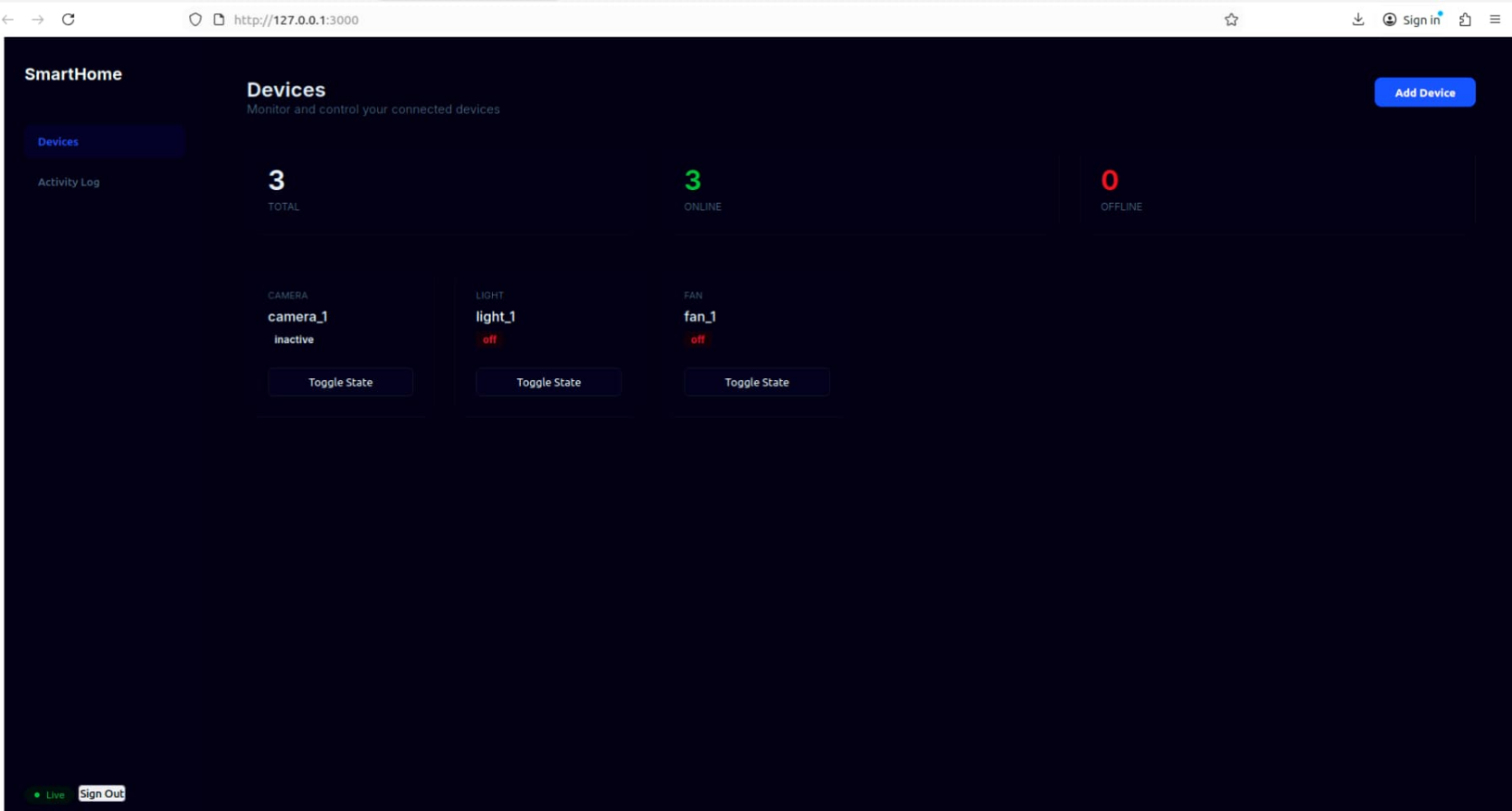
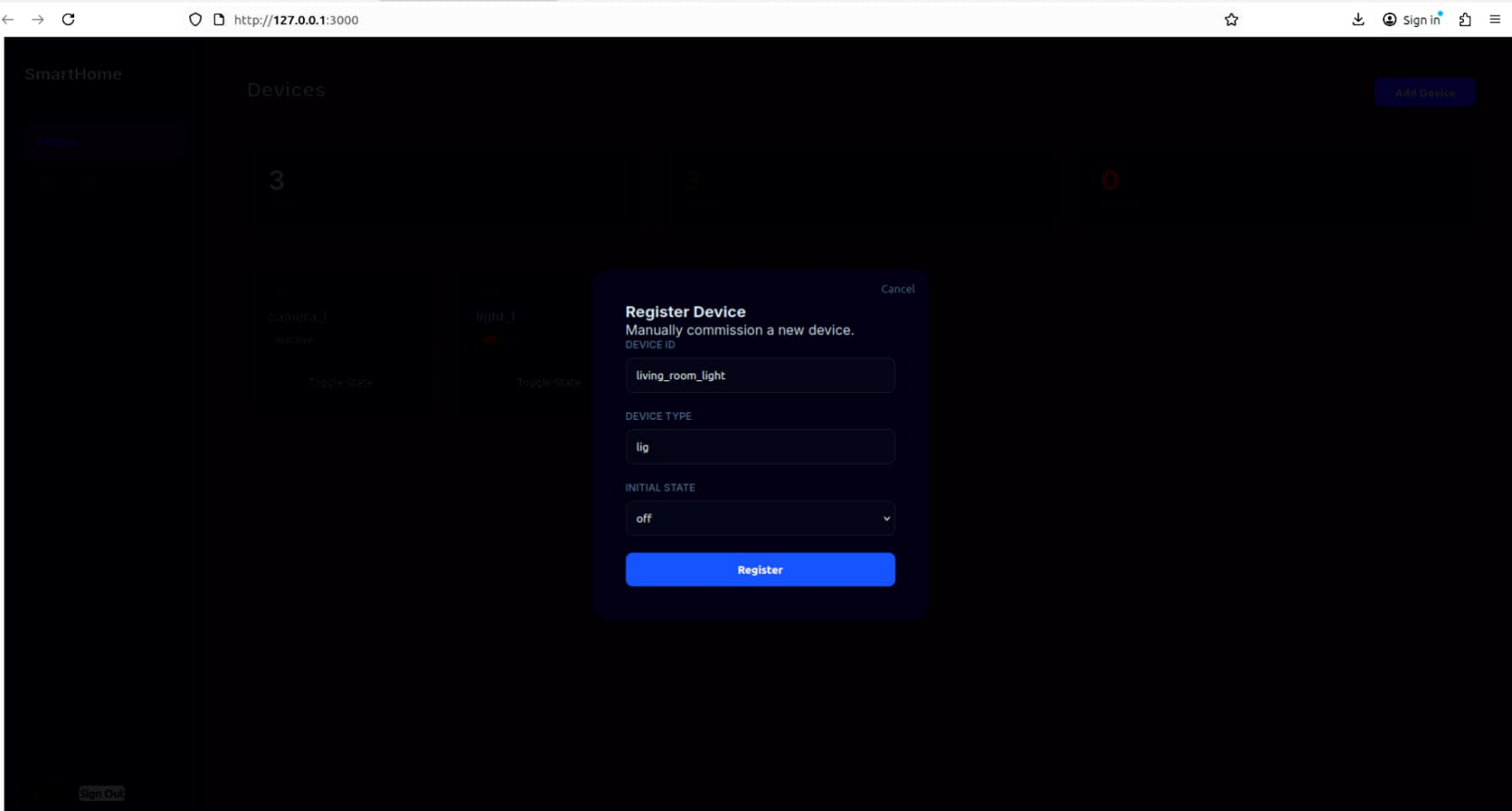
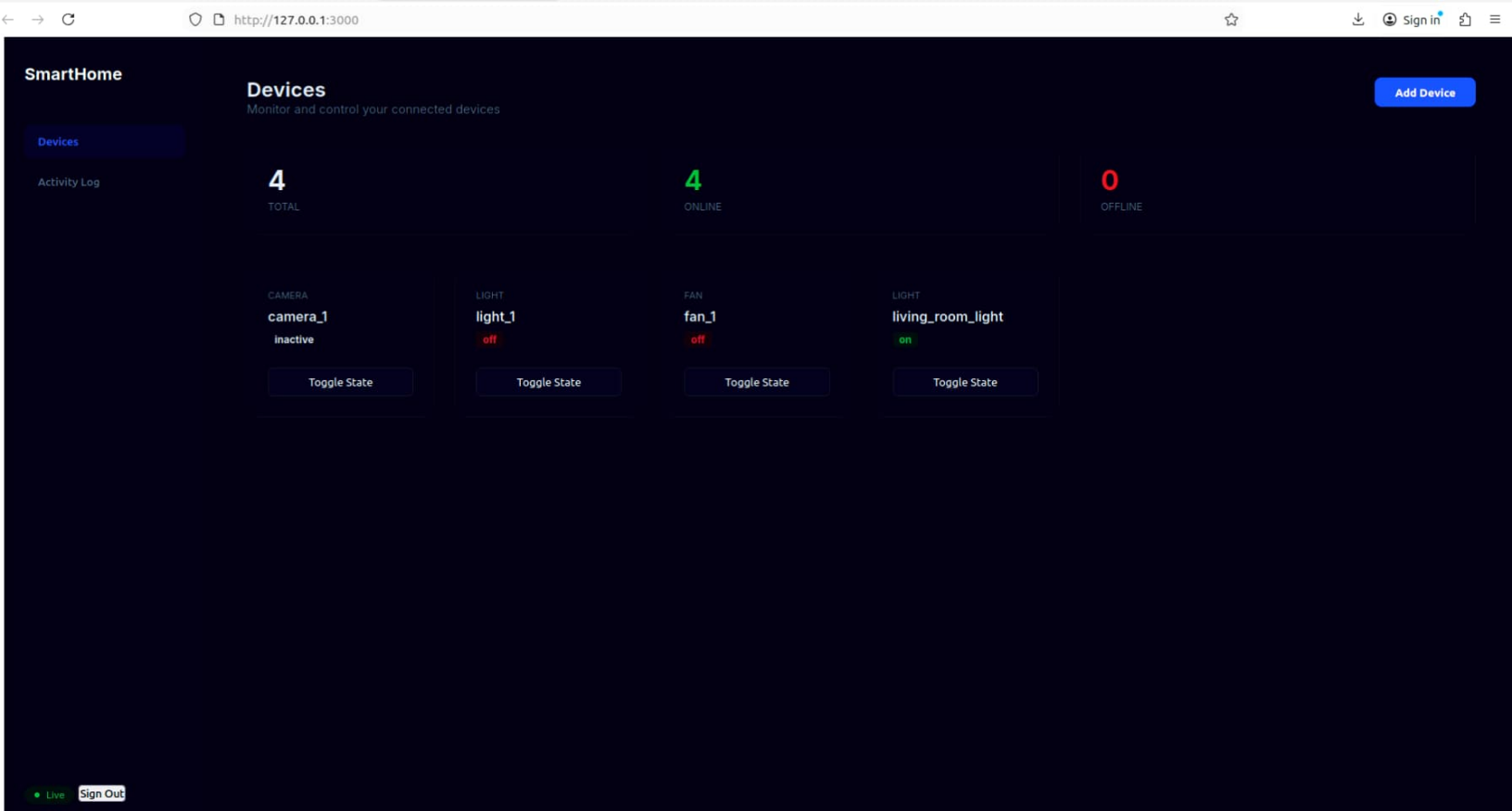

http://127.0.0.1:3000
Sign in
SmartHome
Controller Dashboard
Live
USERNAME
admin
PASSWORD
Sign In


http://127.0.0.1:3000
Sign in
SmartHome
Devices
Activity Log
Devices
Monitor and control your connected devices
Add Device
3
TOTAL
3
ONLINE
0
OFFLINE
CAMERA
camera_1
inactive
Toggle State
LIGHT
light_1
off
Toggle State
FAN
fan_1
off
Toggle State
Live
Sign Out


http://127.0.0.1:3000
Sign in
SmartHome
Devices
Add Device
3
3
0
Cancel
Register Device
Manually commission a new device.
DEVICE ID
living_room_light
DEVICE TYPE
lig
INITIAL STATE
off
Register
Sign Out


http://127.0.0.1:3000
Sign in
SmartHome
Devices
Activity Log
Devices
Monitor and control your connected devices
Add Device
4
TOTAL
4
ONLINE
0
OFFLINE
CAMERA
camera_1
inactive
Toggle State
LIGHT
light_1
off
Toggle State
FAN
fan_1
off
Toggle State
LIGHT
living_room_light
on
Toggle State
Live
Sign Out

Deepam Ahuja 23095316O Dupb

1. Comms. Arch, TCP Socket lifecycle & Concurrency Model.

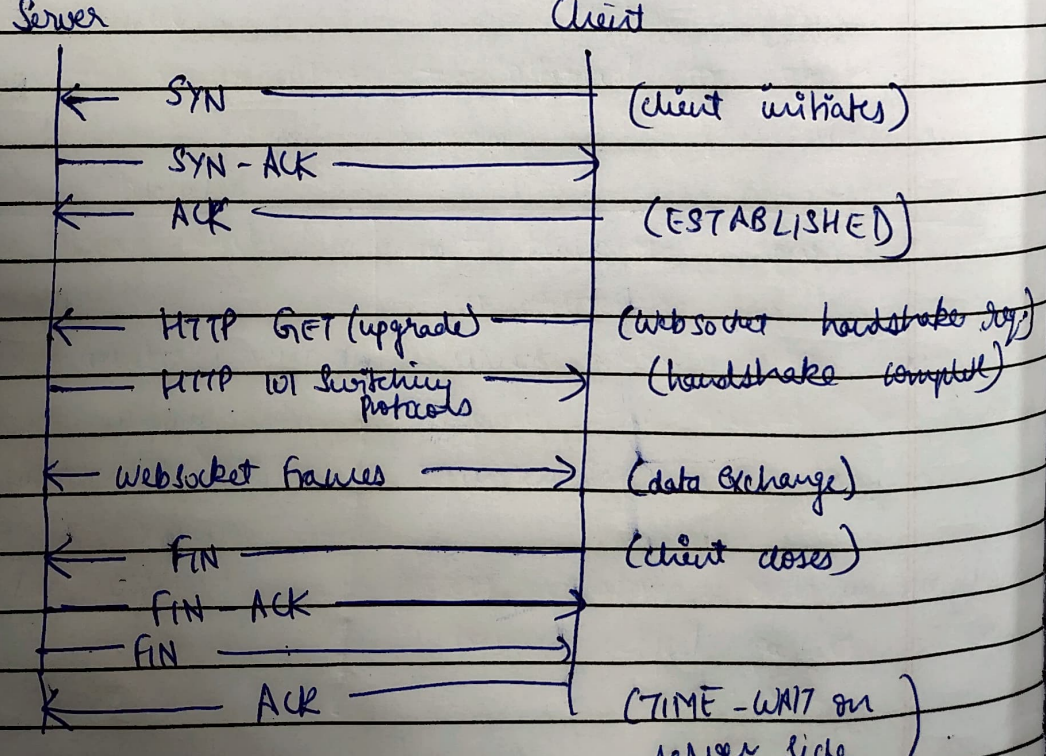
a. System overview

Smart home controller connects 3 types of participants over single TCP to websocket server

Web Browser → Websocket (ws://)
iot device 1 → websocket (ws://)
iot device 2 → websocket (ws://)
→ [C, websocket server : 8080]
|
[SQLiteDB]
[shared memory]

- websocket clients send login, get-device, command, and get-log msg.
- iot device clients send register, heartbeat & state-update msg
- server routes command from web clients to the correct device child process using Posix shared memory + named semaphore + SIGUSR1 signal

b. TCP socket lifecycle.



Deepam Ahuja 230953160

classmate Date Page

TCP socket opt used in server.

- SO-reuseaddr → used in server socket → allows rebinding port 8080 even if it is in TIME-WAIT state after a restart
- TCP-NODELAY → each accepted client socket → disable Nagle's algorithm → commands are sent immediately w/o buffer to reduce latency
- SO-KEEPALIVE → each accepted client socket → OS sends keepalive probes to detect dead connection & clean up close wait state

Code ref (src/main.c)

```
int opt = 1
setsockopt (server_fd, sol_socket, so_reuseaddr, &opt, sizeof(opt));
int flag = 1;
setsockopt (client_fd, ipproto_tcp, tcp_nodelay, &flag, sizeof(flag));
setsockopt (client_fd, sol_socket, so_keepalive, &flag, sizeof(flag));
```

2. Websocket handshake mechanism (RFC 6455)
→ websocket begins as an HTTP/1.1 upgrade req., full RFC 6455 handshake implementation:

i. Send HTTP req by client → GET / HTTP/1.1
Host: localhost:8080
upgrade: websocket
connection: Upgrade
Sec-Websocket-Key: dGhl.....
Sec-websocket-ver: 13

ii. Server computes accept key
↓
accept-key = Base64 (SHA-1 (client key + "258EAFA.....-C5A-B11"))

iii) Server replies with HTTP 101:

```
deep@deep-ASUS-TUF-Gaming-A15-FA506QM-FA506QM:~/Videos/Sem-6-Labs/NPACN_Fisac$ python3 simulator/device sim.py --port 8080
    t.join()
  File "/usr/lib/python3.10/threading.py", line 1096, in join
    self._wait_for_tstate_lock()
  File "/usr/lib/python3.10/threading.py", line 1116, in _wait_for_tstate_lock
    if lock.acquire(block, timeout):
KeyboardInterrupt
^CException ignored in: <module 'threading' from '/usr/lib/python3.10/threading.py'>
Traceback (most recent call last):
  File "/usr/lib/python3.10/threading.py", line 1567, in _shutdown
    lock.acquire()
KeyboardInterrupt:

deep@deep-ASUS-TUF-Gaming-A15-FA506QM-FA506QM:~/Videos/Sem-6-Labs/NPACN_Fisac$ netstat -ant | grep 8080
tcp        0      0 127.0.0.1:47856         127.0.0.1:8080          TIME_WAIT
tcp        0      0 127.0.0.1:8080          127.0.0.1:52884         ESTABLISHED
tcp        0      0 127.0.0.1:52884         127.0.0.1:8080          ESTABLISHED
tcp        0      0 127.0.0.1:47866         127.0.0.1:8080          TIME_WAIT
tcp        0      0 127.0.0.1:47840         127.0.0.1:8080          TIME_WAIT
deep@deep-ASUS-TUF-Gaming-A15-FA506QM-FA506QM:~/Videos/Sem-6-Labs/NPACN_Fisac$
```

bash
bash server

⊗ 0 ⚠ 0

Deepanshu Ahuja 230953160

Def

HTTP/1.1 101 switching protocol.
upgrade: websocket
Connection: upgrade
sec-websocket-accept : aZBd...

After this exchange, TCP connection is upgraded, both sides using websocket framing protocol instead of HTTP.

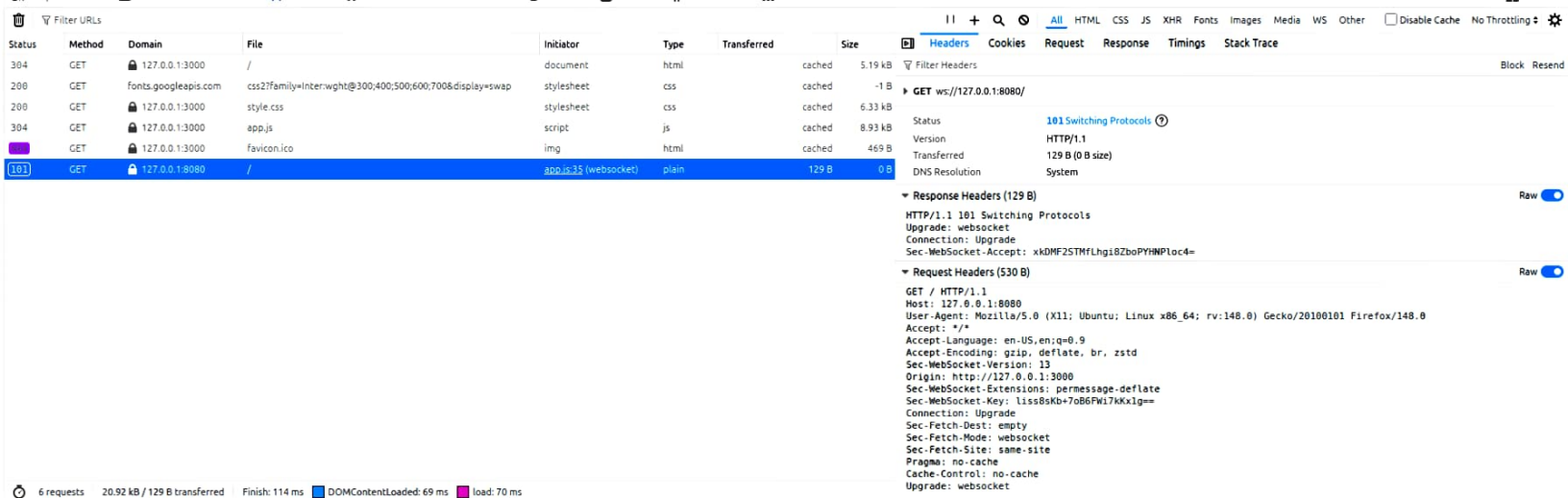
Code ref (src/websocket.c)

```
SHA1 ((unsigned char *) combined, strlen(combined), sha1_res);
char *acc-key = base64_encode(sha1_res, 20);
snprintf (response, sizeof(response),
        "HTTP/1.1 101 switching Protocol\r\n"
        "upgrade : websocket \r\n"
        "connection: upgrade \r\n"
        "sec-websocket-accept : %s\r\n\r\n", accept_key)
```

d. Concurrency model
- ↳ chose fork() per connection + pthread for bg tasks.

fork() alt & justification:

→ single thread select()	can't block on recv() w/o starving other connections.
→ thread/connection (only pthread)	shared address space means new bug in one thread crash everything.
→ async i/o (epoll)	significantly more complex.
→ fork() per connection	process isolation; crash in one child doesn't affect parent & other connections.


Filter URLs
All HTML CSS JS XHR Fonts Images Media WS Other
Disable Cache
No Throttling
Status Method Domain File Initiator Type Transferred Size
304 GET 127.0.0.1:3000 / document html cached 5.19 kB
200 GET fonts.googleapis.com css2?family=Inter:wght@300;400;500;600;700&display=swap stylesheet css cached -1 B
200 GET 127.0.0.1:3000 style.css stylesheet css cached 6.33 kB
304 GET 127.0.0.1:3000 app.js script js cached 8.93 kB
GET 127.0.0.1:3000 favicon.ico img html cached 469 B
101 GET 127.0.0.1:8080 / app.js:35 (websocket) plain 129 B 0 B
6 requests 20.92 kB / 129 B transferred Finish: 114 ms DOMContentLoaded: 69 ms load: 70 ms
Headers Cookies Request Response Timings Stack Trace
Filter Headers
Block Resend
GET ws://127.0.0.1:8080/
Status 101 Switching Protocols
Version HTTP/1.1
Transferred 129 B (0 B size)
DNS Resolution System
Response Headers (129 B)
Raw
HTTP/1.1 101 Switching Protocols
Upgrade: websocket
Connection: Upgrade
Sec-WebSocket-Accept: xkDMF2STMfLhgi8ZboPYHNPloc4=
Request Headers (530 B)
Raw
GET / HTTP/1.1
Host: 127.0.0.1:8080
User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:148.0) Gecko/20100101 Firefox/148.0
Accept: */*
Accept-Language: en-US,en;q=0.9
Accept-Encoding: gzip, deflate, br, zstd
Sec-WebSocket-Version: 13
Origin: http://127.0.0.1:3000
Sec-WebSocket-Extensions: permessage-deflate
Sec-WebSocket-Key: liss8sKb+7oB6FWi7kKx1g==
Connection: Upgrade
Sec-Fetch-Dest: empty
Sec-Fetch-Mode: websocket
Sec-Fetch-Site: same-site
Pragma: no-cache
Cache-Control: no-cache
Upgrade: websocket

23095360

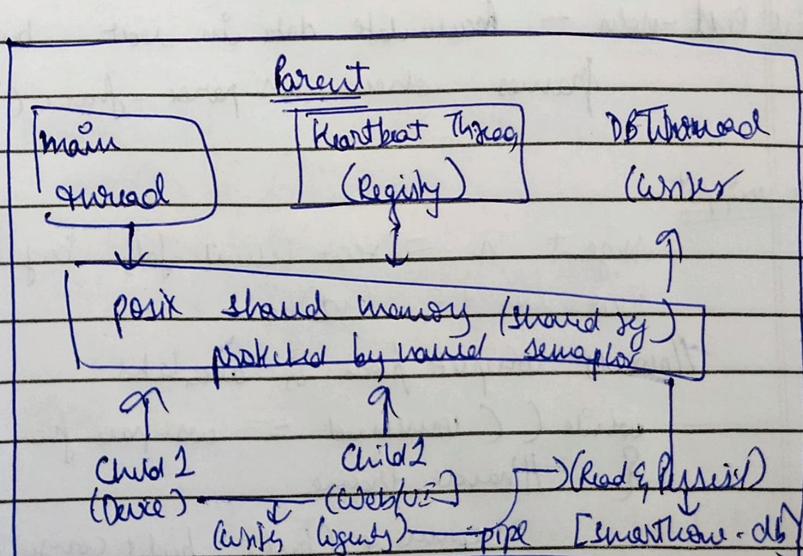
Thread responsibility.

- main thread → accept loop → for new TCP connection, fork child / client.
- db-write-thread → DB writer → drain log pipe to write to sqlite — serialize all log entry.
- heartbeat-thread → heartbeat monitor → every 30s, scans shared memory & mark device offline if no heartbeat.

IPC b/w parent & children

Parent process — child process (per device)

- Posix shared memory (Shared Registry) ↔ mutex via named semaphore
- Pipe (db-pipe) ← child writes log entry there
- SIGUSER1 → child wakes up, reads pending commands



Parent [SIGUSR2] (Send command notify) → Child (Device)

Inspector | Console | Debugger | Network | Style Editor | Performance | Memory | Storage | Accessibility | Application

Filter URLs

All | HTML | CSS | JS | XHR | Fonts | Images | Media | WS | Other | Disable Cache | No Throttling

Status	Method	Domain	File	Initiator	Type	Transferred	Size	0 ms / 10.24 s / 20.48 s
304	GET	127.0.0.1:3000	/	document	html	cached	5.19 kB	5 ms
200	GET	fonts.googleapis.com	css2?family=Inter:wght@300;400;500;600;700&display=swap	stylesheet	css	cached	-1 B	0 ms
200	GET	127.0.0.1:3000	style.css	stylesheet	css	cached	6.33 kB	0 ms
304	GET	127.0.0.1:3000	app.js	script	js	cached	8.93 kB	1 ms
	GET	127.0.0.1:8080	/	app.js:35 (websocket)		NS_ERROR_CONNECTION_REFUSED	0 B	0 ms
[illegible]	GET	127.0.0.1:3000	favicon.ico	img	html	cached	469 B	0 ms
	GET	127.0.0.1:8080	/	app.js:35 (websocket)		NS_ERROR_CONNECTION_REFUSED	0 B	0 ms
	GET	127.0.0.1:8080	/	app.js:35 (websocket)		NS_ERROR_CONNECTION_REFUSED	0 B	0 ms
	GET	127.0.0.1:8080	/	app.js:35 (websocket)		NS_ERROR_CONNECTION_REFUSED	0 B	0 ms
	GET	127.0.0.1:8080	/	app.js:35 (websocket)		NS_ERROR_CONNECTION_REFUSED	0 B	0 ms
	GET	127.0.0.1:8080	/	app.js:35 (websocket)		NS_ERROR_CONNECTION_REFUSED	0 B	0 ms
	GET	127.0.0.1:8080	/	app.js:35 (websocket)		NS_ERROR_CONNECTION_REFUSED	0 B	0 ms
101	GET	127.0.0.1:8080	/	app.js:35 (websocket)	plain	129 B	0 B	1 ms

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Q1

2. Websocket server implementation details.

a) handling multiple client types
   ↳ done through 1st msg

TCP connect → WS handshake → first msg:
   * type = auth → client-web (browser dashboard)
   type = register → client-device (IoT device)

Each child process holds a 'Connection' struct that tracks the client's type, state, session token, device ID and a recv-buffer for partial reads.

b. Partial read & write handling.

Problem: tcp is a stream protocol, single recv() call may return less data than one complete WebSocket frame.
send() / write() may send only part of frame.

Partial Read → soln – Accumulate data in recv_buf and only process frames when ws_parse_frame() returns the value.

code snippet

```
ssize_t n = recv(client_fd, buf + recv_len, sizeof(buf) - recv_len, 0);
recv_len += (int)n;
//process complete frame is available
while ((consumed = ws_parse_frame(buf, recv_len, ...)) > 0)
{   //handle frame
    memmove(buf, buf + consumed, recv_len - consumed);
    recv_len -= consumed;
}
```

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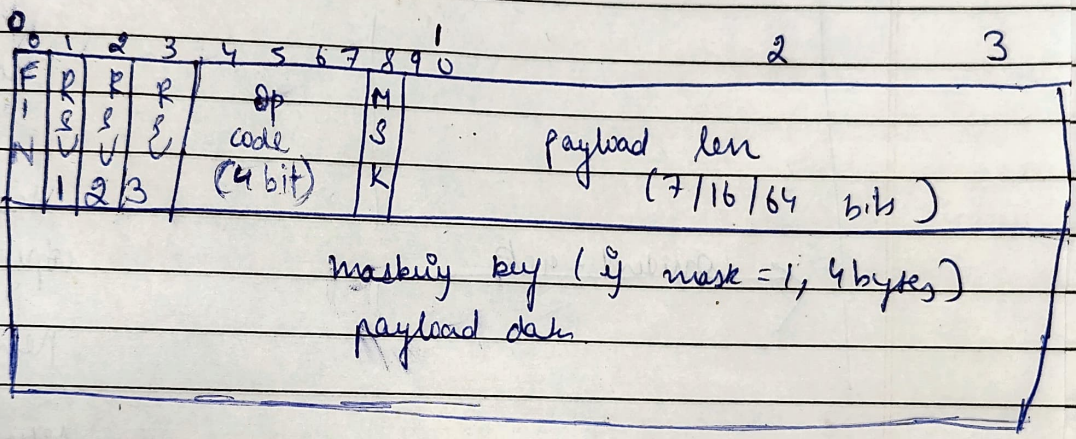
Partial writes

ws_send_frame() loops until all bytes are written.

code snippet

```
while (total < frame_len) {
    ssize_t n = write(fd, frame + total, frame_len - total);
    if (n < 0 && (errno == EAGAIN)) {
        usleep(1000); continue;
    }
    total += (uint)n;
}
```

Websocket frame format.



frame parser handles all 3 payload len encoding:

<126 → 7 bit len in byte 1

=126 → next 2 bytes give len

=127 → next 8 bytes give length.

opcode handled.

opcode	Hex	Action
text frame	0x1	route to hand_message()
binary frame	0x2	reject with close code 1003
close	0x8	echo close frame, exit child
ping	0x9	respond with pong (same payload)
pong	0xA	silent ignore
oversize (>64KB)	—	close with code 1009

```
deep@deep-ASUS-TUF-Gaming-A15-FA506QM-FA506QM:~/Videos/Sem-6-Labs/NPACN_Fisac$ python3 simulator/device_sim.py --port 8080
[fan_1] Heartbeat sent
[light_1] Heartbeat sent
[fan_1] Heartbeat sent
[camera_1] Heartbeat sent
[light_1] Heartbeat sent
[light_1] Heartbeat sent
[fan_1] Heartbeat sent
[camera_1] Heartbeat sent
[light_1] Heartbeat sent
[camera_1] Heartbeat sent
[fan_1] Heartbeat sent
[light_1] Heartbeat sent
[camera_1] Heartbeat sent
[fan_1] Heartbeat sent
[camera_1] Heartbeat sent
[light_1] Heartbeat sent
[fan_1] Heartbeat sent
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[light_1] Heartbeat sent
[camera_1] Heartbeat sent
[fan_1] Heartbeat sent
[light_1] Heartbeat sent
[camera_1] Heartbeat sent
[fan_1] Heartbeat sent
[camera_1] Heartbeat sent
[light_1] Heartbeat sent
[fan_1] Heartbeat sent
[camera_1] Heartbeat sent
```

```
deep@deep-ASUS-TUF-Gaming-A15-FA506QM-FA506QM:~/Videos/Sem-6-Labs/NPACN_Fisac$ cd client
deep@deep-ASUS-TUF-Gaming-A15-FA506QM-FA506QM:~/Videos/Sem-6-Labs/NPACN_Fisac/client$ python3 -m http.server 3000
Serving HTTP on 0.0.0.0 port 3000 (http://0.0.0.0:3000/) ...
127.0.0.1 - - [31/Mar/2026 16:41:44] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [31/Mar/2026 16:41:44] "GET /style.css HTTP/1.1" 200 -
127.0.0.1 - - [31/Mar/2026 16:41:44] "GET /app.js HTTP/1.1" 200 -
127.0.0.1 - - [31/Mar/2026 16:41:44] code 404, message File not found
127.0.0.1 - - [31/Mar/2026 16:41:44] "GET /favicon.ico HTTP/1.1" 404 -
```

```
deep@deep-ASUS-TUF-Gaming-A15-FA506QM-FA506QM:~/Videos/Sem-6-Labs/NPACN_Fisac/server$ ./server 8080
[INFO] Database initialized
[INFO] Server listening on port 8080 using database smarthome.db
[INFO] Child 23878 handling connection from 127.0.0.1:36286
[INFO] WebSocket handshake successful in child
[INFO] Auth successful
[INFO] Child 23910 handling connection from 127.0.0.1:51412
[INFO] Child 23911 handling connection from 127.0.0.1:51414
[INFO] Child 23912 handling connection from 127.0.0.1:51418
[INFO] WebSocket handshake successful in child
[INFO] WebSocket handshake successful in child
[INFO] WebSocket handshake successful in child
[INFO] Device registered
[INFO] Device registered
[INFO] Device registered
[INFO] Child 24266 handling connection from 127.0.0.1:50758
[INFO] Child 24267 handling connection from 127.0.0.1:50772
[INFO] Child 24268 handling connection from 127.0.0.1:50788
[INFO] WebSocket handshake successful in child
[INFO] WebSocket handshake successful in child
[INFO] WebSocket handshake successful in child
[INFO] Device registered
[INFO] Device registered
[INFO] Device registered
[INFO] Child 24689 handling connection from 127.0.0.1:39752
[INFO] WebSocket handshake successful in child
[INFO] Auth successful
[INFO] Command received for device
[INFO] Command received for device
```

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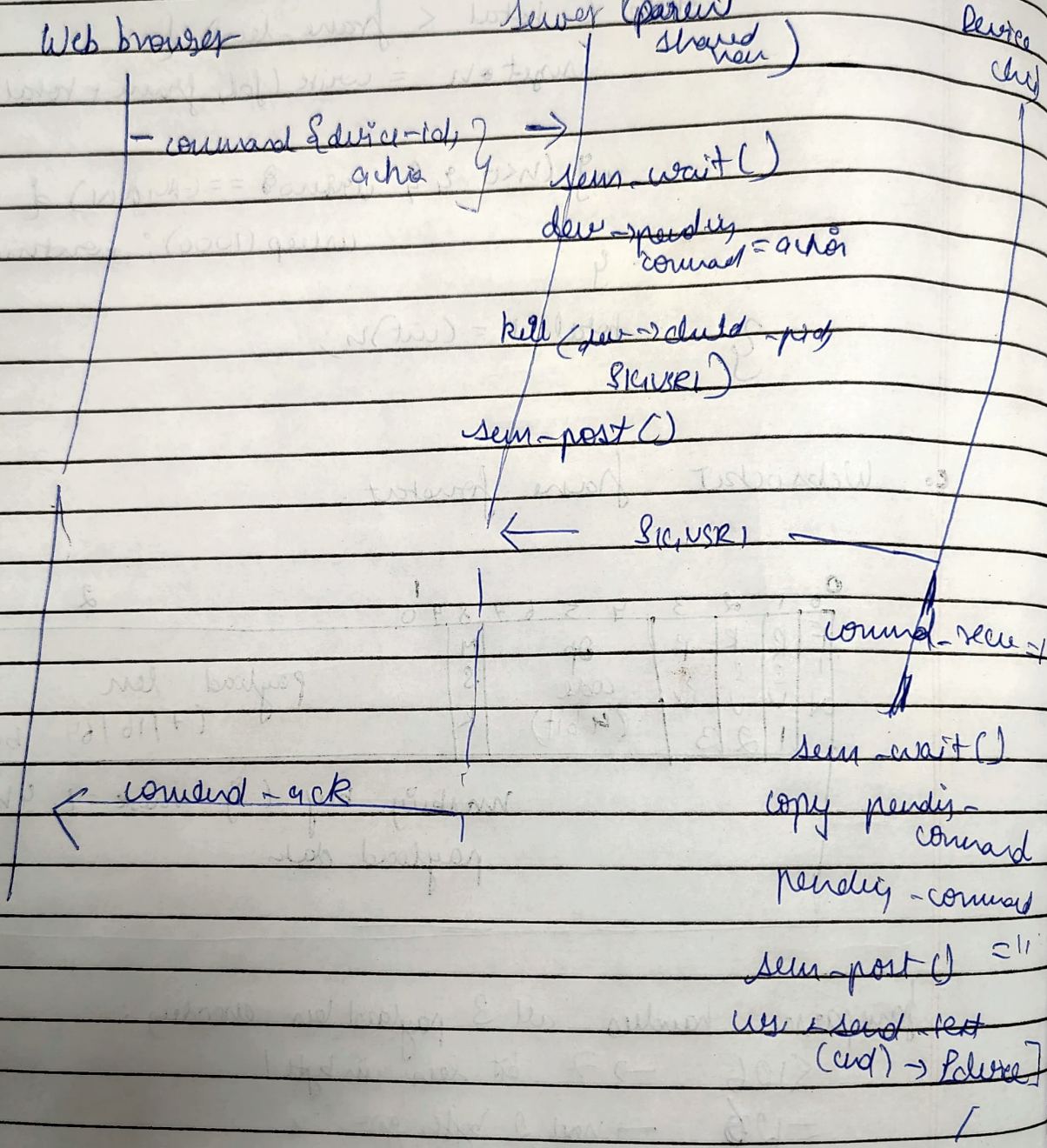
d. command path (web → device)



Kotha Rushed Reddy 230953176 CCE-C-24

Schema Design

Users Tables used stores admin credentials:

```
Create TABLE IF NOT EXISTS users (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    Username TEXT UNIQUE NOT NULL
    Password_hash TEXT NOT NULL,
    created_at INTEGER NOT NULL
);
```

devices table - Persistent device registry:

```
Create TABLE IF NOT EXISTS devices(
    device_id TEXT PRIMARY KEY,
    device_type TEXT NOT NULL,
    State       TEXT NOT NULL DEFAULT 'offline',
    last_seen   Integer NOT NULL DEFAULT 0
);
```

activity log table - command audit trail:

```
CREATE TABLE IF NOT EXISTS activity_log (
    id Integer PRIMARY KEY AUTOINCRIMENT,
    timestamp INTEGER NOT NULL,
    username TEXT NOT NULL DEFAULT '',
    device_id TEXT NOT NULL DEFAULT '',
    action    TEXT NOT NULL,
    source_ip TEXT NOT NULL DEFAULT ''
);
```

server > smarthome.db

Tables

Search tables...

- activity_log 13
- > devices 4
- sqlite_sequence 2
- users 1

devices

Refresh | Schema | Export CSV | Export JSON | Copy All | Visualize

SQL Query Editor (click to expand)

Search...

4 total rows | 4 columns

ROWID	DEVICE_ID	DEVICE_TYPE	STATE	LAST_SEEN
4	koi_pond_lights	light	on	1774930093
11	camera_1	camera	inactive	1774947397
12	light_1	light	off	1774947397
13	fan_1	fan	off	1774947387

server > smarthome.db

Tables

Search tables...

- activity_log 13
- devices 5
- sqlite_sequence 2
- users 1

sqlite_sequence

Refresh | Schema | Export CSV | Export JSON | Copy All | Visualize

SQL Query Editor (click to expand)

Search...

2 total rows | 2 columns

ROWID	NAME	SEQ
1	users	1
2	activity_log	13

server > smarthome.db

Tables

Search tables...

- activity_log 17
- devices 5
- sqlite_sequence 2
- > users 1

users

Refresh | Schema | Export CSV | Export JSON | Copy All | Visualize

SQL Query Editor (click to expand)

Search...

1 total rows | 4 columns

ID	USERNAME	PASSWORD_HASH	CREATED_AT
1	admin	240be518fabd2724ddb6f04eeb1da5967448d7e831c08c8fa822809f74c720a9	1774929933

server > smarthome.db

Tables

Search tables...

- activity_log 13
- devices
- sqlite_sequence
- users

activity_log

Refresh | Schema | Export CSV | Export JSON | Copy All | Visualize

SQL Query Editor (click to expand)

Search...

13 total rows | 6 columns

ID	TIMESTAMP	USERNAME	DEVICE_ID	ACTION	SOURCE_IP
1	1774930056	admin	camera_1	active	127.0.0.1
2	1774930058	admin	fan_1	on	127.0.0.1
3	1774930096	admin	fan_1	off	127.0.0.1
4	1774930196	admin	fan_1	on	127.0.0.1
5	1774930198	admin	light_1	on	127.0.0.1
6	1774930199	admin	camera_1	inactive	127.0.0.1

Kotha Rishul Reddy 230953176 CLE-C-24

Prepared Statements

All queries are compiled once at startup as sqlite3_stmt* pointers and reused with sqlite3_bind_* / sqlite3_reset(). This prevents SQL injection and avoids per-query compilation cost.

Six Prepared statements

Variable	SQL Purpose
stmt_user_lookup	Fetch password hash by name
stmt_device_insert	insert or replace device row
stmt_device_update	update device state + last_seen
stmt_device_list	Fetch all devices (control)
stmt_log_insert	insert activity log row
stmt_log_fetch	Fetch last N log rows

Database Synchronization & Real-Time Impact

Problem: Multi child processes can call db_update_device concurrently.

Solution: WAL mode:

PRAGMA journal_mode = WAL;

PRAGMA synchronous = NORMAL;

Activity log write path - serialised via Pipe:

All log writes go through a pipe to the parents db_write-thread, which is the only thread that calls db_log_activity. This eliminates writer concurrent writer contention for most frequent write path

__/__/____

Kotha Rushed Reddy 2309453176 CCE-C-24

Real time Input Analysis

Operation	Frequency	Latency	Impact
db_log_activity	Every command	~0.2ms	LOW
db_update_device	Every heartbeat(30s)	~ 0.3ms	Negligible
db_get_activity_log	Every 5s(poll)	~ 1ms	Acceptable

Authoritative real-time source: The in-memory Shared Registry, NOT the database. The DB is used only for persistence and log queries.

Activity Log

Last 20 events across all devices

Time	User	Device	Action
2:29:25 PM	admin	camera_1	active
2:29:25 PM	admin	light_1	on
2:29:20 PM	admin	fan_1	on
2:29:20 PM	admin	fan_1	on
2:23:37 PM	admin	camera_1	active
9:48:22 AM	admin	light_1	on
9:46:32 AM	admin	light_1	off
9:46:28 AM	admin	camera_1	active
9:46:27 AM	admin	fan_1	off
9:46:26 AM	admin	fan_1	on
9:46:24 AM	admin	light_1	on
9:39:59 AM	admin	camera_1	inactive
9:39:58 AM	admin	light_1	on
9:39:56 AM	admin	fan_1	on
9:38:16 AM	admin	fan_1	off
9:37:38 AM	admin	fan_1	on

Tanish Sam Nathan
230953192

Authentication and TCP State Impact:

Authentication Flow:

->Step1 - Password hashing (SHA-256): Passwords are never stored in plain text. On login, the input password is hashed with SHA-256 using OpenSSL and compared against the stored hash:

```
char * auth_hash_password(const char *password){
    unsigned char hash[SHA256_DIGEST_LENGTH];
    SHA256((unsigned char*) password, strlen(password),
        hash);
}
```

Default seed: username admin, password admin123 ->
SHA-256: 240be518fabd.....

->Step2 - Session Token generation: On successful login, 32 random bytes are read from /dev/urandom and hex-encoded into a 64-character token:

```
void auth_generate_token(char *out_token){
    int fd = open("/dev/urandom", O_RDONLY);
    read(fd, buf, 32);
}
```

->Step3 - Token validation on every subsequent message:

Every get_devices, command, and get_log message must include token. The server calls auth_validate_token() before processing.

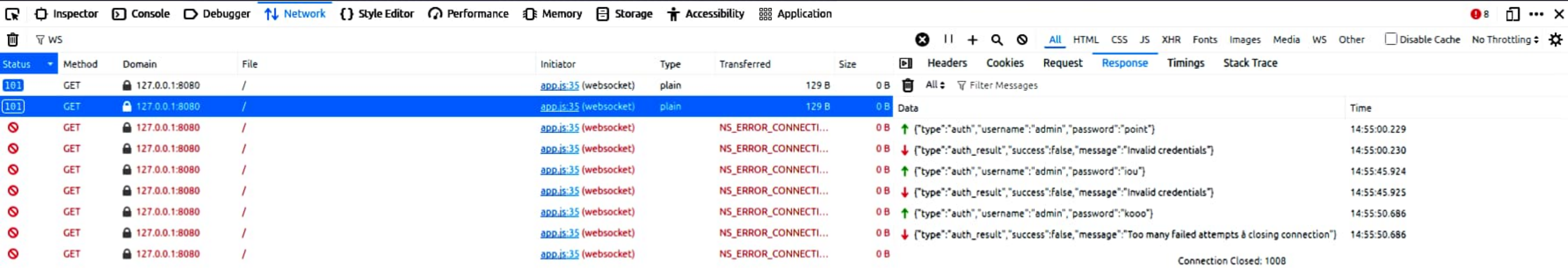
```
int auth_validate_token(const Connection *conn, const char
                        char* token) {
    return (strncmp(conn->session_token, token, 64)
            == 0) ? 1 : 0;
}
```

3 - Attempt lockout: After 3 consecutive failed login attempts, the server sends a final error message, closes the WebSocket connection with code 1008 (Policy Violation), and sets a close_conn flag to break the child process loop.

* WebSocket Security Design:

-> Our implementation uses plain ws:// (no TLS) as required by the project spec. In a production deployment, wss:// (WebSocket Secure over TLS) would be used — this encrypts the entire connection so that the tokens and commands cannot be intercepted.

why ws:// is acceptable:
-> Server runs only on localhost
-> Network traffic does not traverse public infrastructure
-> The SHA-256 password hash still protects credentials at rest in DB


Inspector
Console
Debugger
Network
Style Editor
Performance
Memory
Storage
Accessibility
Application
WS
All
HTML
CSS
JS
XHR
Fonts
Images
Media
WS
Other
Disable Cache
No Throttling
Status
Method
Domain
File
Initiator
Type
Transferred
Size
101 GET 127.0.0.1:8080 / app.js:35 (websocket) plain 129 B 0 B
101 GET 127.0.0.1:8080 / app.js:35 (websocket) plain 129 B 0 B
GET 127.0.0.1:8080 / app.js:35 (websocket) NS_ERROR_CONNECTI... 0 B
GET 127.0.0.1:8080 / app.js:35 (websocket) NS_ERROR_CONNECTI... 0 B
GET 127.0.0.1:8080 / app.js:35 (websocket) NS_ERROR_CONNECTI... 0 B
GET 127.0.0.1:8080 / app.js:35 (websocket) NS_ERROR_CONNECTI... 0 B
GET 127.0.0.1:8080 / app.js:35 (websocket) NS_ERROR_CONNECTI... 0 B
GET 127.0.0.1:8080 / app.js:35 (websocket) NS_ERROR_CONNECTI... 0 B
GET 127.0.0.1:8080 / app.js:35 (websocket) NS_ERROR_CONNECTI... 0 B
Headers
Cookies
Request
Response
Timings
Stack Trace
All
Filter Messages
Data
Time
{"type":"auth","username":"admin","password":"point"} 14:55:00.229
{"type":"auth_result","success":false,"message":"Invalid credentials"} 14:55:00.230
{"type":"auth","username":"admin","password":"iou"} 14:55:45.924
{"type":"auth_result","success":false,"message":"Invalid credentials"} 14:55:45.925
{"type":"auth","username":"admin","password":"kooo"} 14:55:50.686
{"type":"auth_result","success":false,"message":"Too many failed attempts à closing connection"} 14:55:50.686
Connection Closed: 1008

TCP state Analysis : TIME_WAIT and CLOSE_WAIT

-) TIME_WAIT :

-) occurs on the server side after the server actively closes a connection

-) The socket stays in TIME_WAIT for 2 x MSL to absorb any delayed TCP segments that arrive after the connection closes

-) Impact on our system : If the server restarts without SO_REUSEADDR, bind() fails because port 8680 is occupied by a TIME_WAIT socket.

-) our fix : setsockopt(server_fd, SOL_SOCKET, SO_REUSEADDR, &opt, sizeof(opt)) - allows immediate re-binding during TIME_WAIT.

-) CLOSE_WAIT :

-) occurs when the remote peer (client/device) sends FIN but our server has not yet closed its end.

-) A large number of CLOSE_WAIT sockets = resource leak (each hold a file descriptor)

-) our fix : SO_KEEPALIVE is set on every accepted socket. The OS sends keepalive probes after ~2 hours of inactivity, if the remote is unreachable, the connection is reset, moving the socket out of CLOSE_WAIT automatically.

-) Additionally, the heartbeat thread marks devices as offline after HEARTBEAT_TIMEOUT seconds using the in-memory registry

TCP State flow for a device disconnect:

Established → (device dies, no FIN)

↓ keepalive probes sent
↓ no response after 9 probes

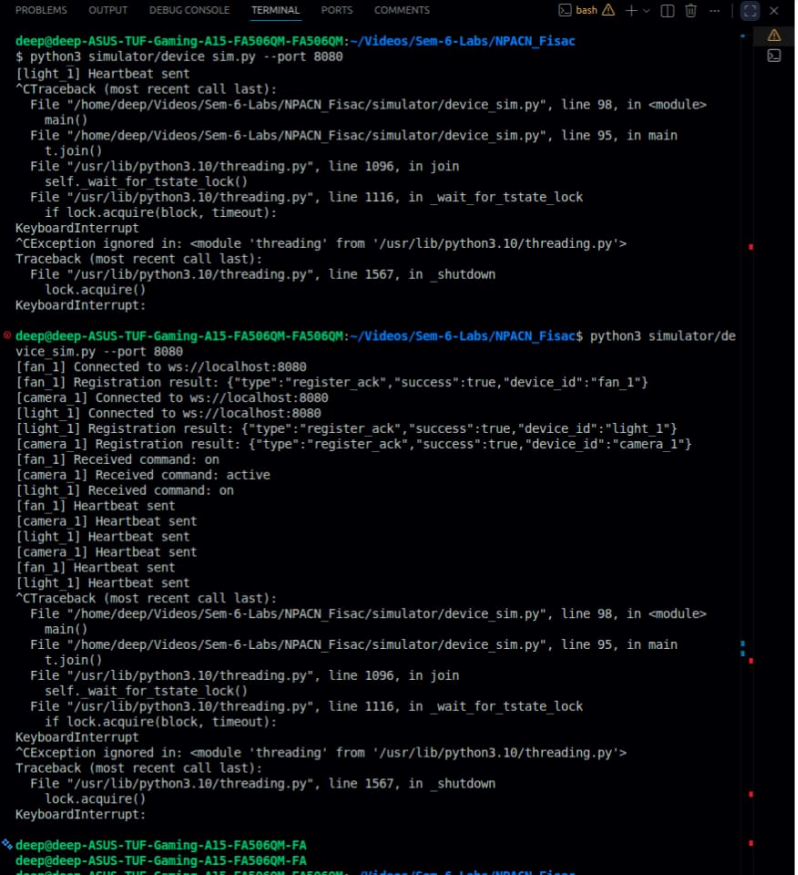
CLOSE_WAIT → kernel resets

↓

CLOSED → child process

↓

dev_set_offline() called → dc


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
COMMENTS
bash
deep@deep-ASUS-TUF-Gaming-A15-FA506QM-FA506QM:~/Videos/Sem-6-Labs/NPACN_Fisac
$ python3 simulator/device_sim.py --port 8080
[light_1] Heartbeat sent
^CTraceback (most recent call last):
File "/home/deep/Videos/Sem-6-Labs/NPACN_Fisac/simulator/device_sim.py", line 98, in <module>
main()
File "/home/deep/Videos/Sem-6-Labs/NPACN_Fisac/simulator/device_sim.py", line 95, in main
t.join()
File "/usr/lib/python3.10/threading.py", line 1096, in join
self._wait_for_tstate_lock()
File "/usr/lib/python3.10/threading.py", line 1116, in _wait_for_tstate_lock
if lock.acquire(block, timeout):
KeyboardInterrupt
^CException ignored in: <module 'threading' from '/usr/lib/python3.10/threading.py'>
Traceback (most recent call last):
File "/usr/lib/python3.10/threading.py", line 1567, in _shutdown
lock.acquire()
KeyboardInterrupt:
deep@deep-ASUS-TUF-Gaming-A15-FA506QM-FA506QM:~/Videos/Sem-6-Labs/NPACN_Fisac$ python3 simulator/device_sim.py --port 8080
[fan_1] Connected to ws://localhost:8080
[fan_1] Registration result: {"type":"register_ack","success":true,"device_id":"fan_1"}
[camera_1] Connected to ws://localhost:8080
[light_1] Connected to ws://localhost:8080
[light_1] Registration result: {"type":"register_ack","success":true,"device_id":"light_1"}
[camera_1] Registration result: {"type":"register_ack","success":true,"device_id":"camera_1"}
[fan_1] Received command: on
[camera_1] Received command: active
[light_1] Received command: on
[fan_1] Heartbeat sent
[camera_1] Heartbeat sent
[light_1] Heartbeat sent
[camera_1] Heartbeat sent
[fan_1] Heartbeat sent
[light_1] Heartbeat sent
^CTraceback (most recent call last):
File "/home/deep/Videos/Sem-6-Labs/NPACN_Fisac/simulator/device_sim.py", line 98, in <module>
main()
File "/home/deep/Videos/Sem-6-Labs/NPACN_Fisac/simulator/device_sim.py", line 95, in main
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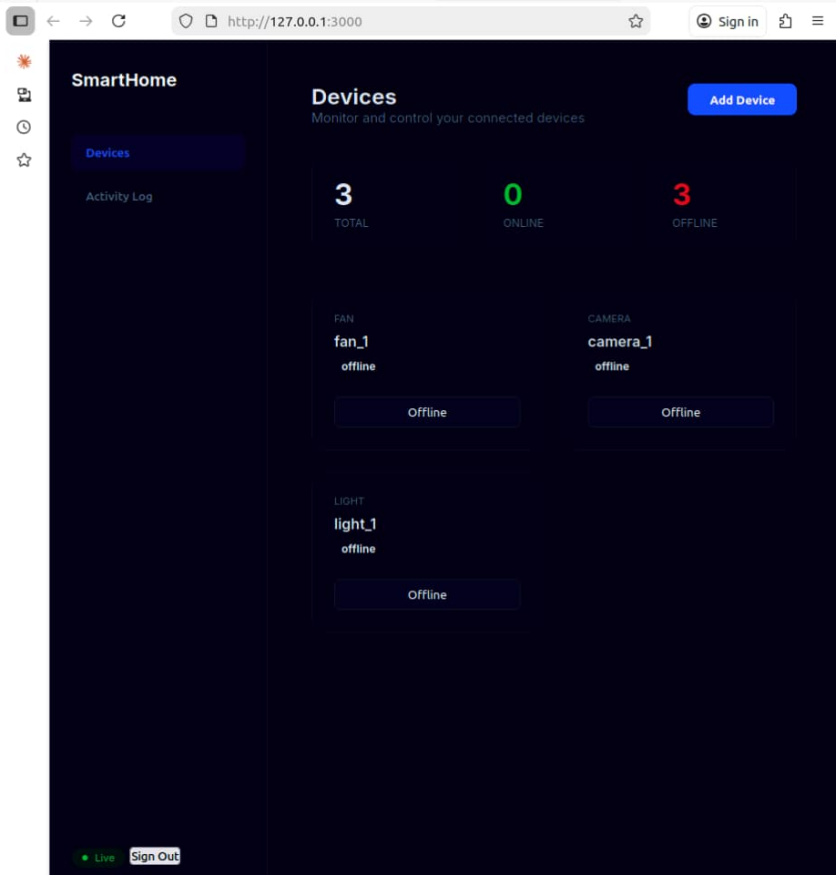
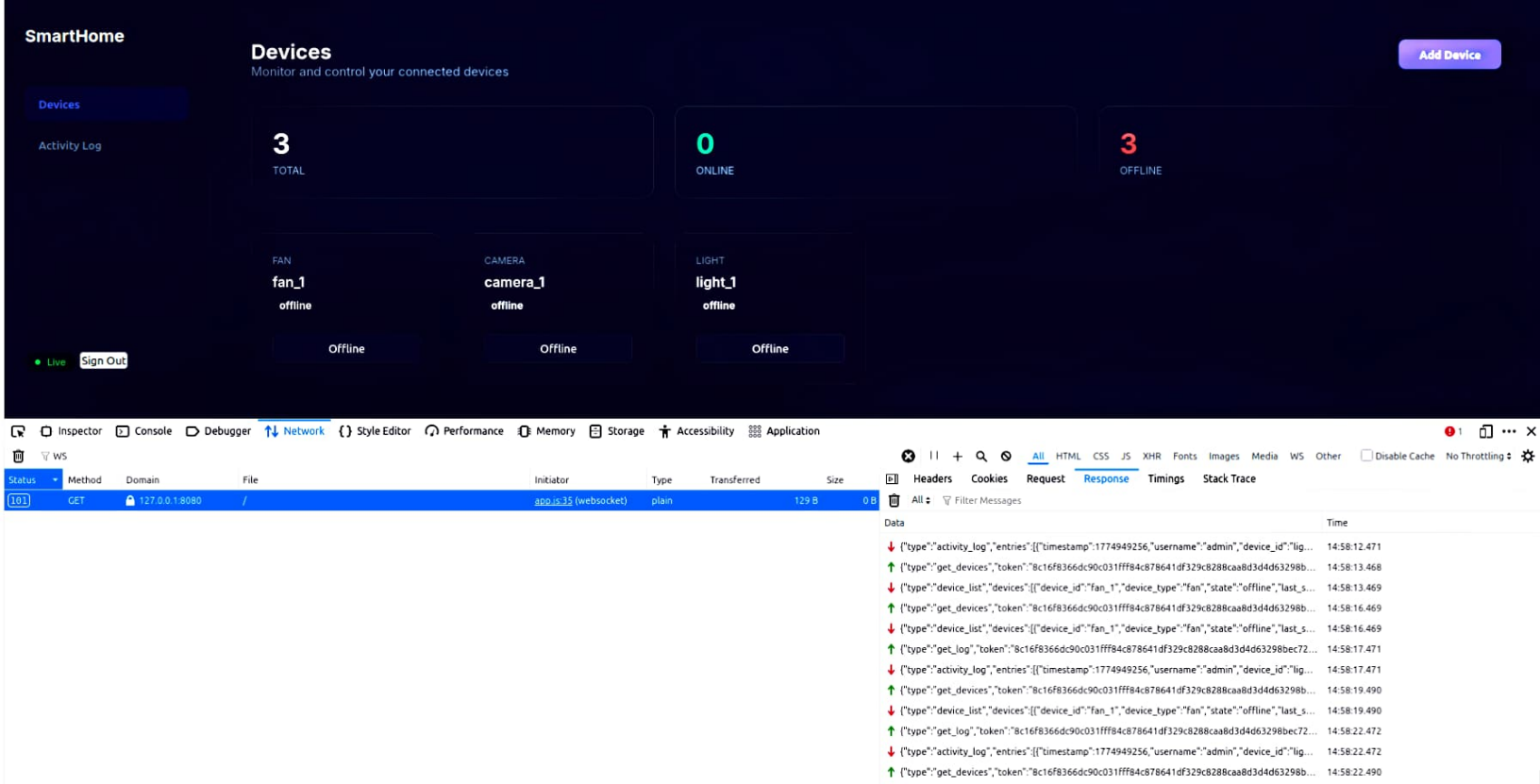
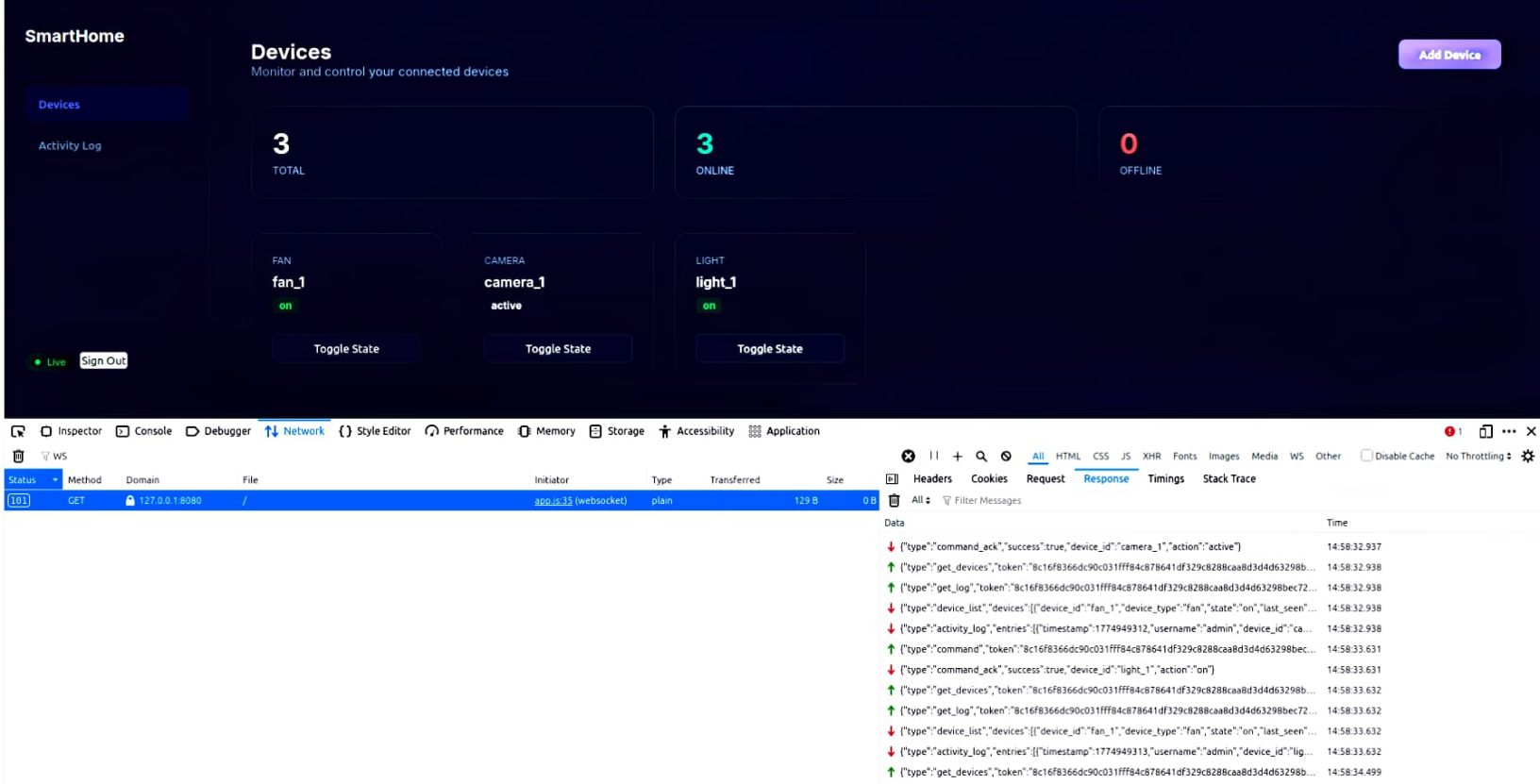
File "/usr/lib/python3.10/threading.py", line 1567, in _shutdown
lock.acquire()
KeyboardInterrupt:
deep@deep-ASUS-TUF-Gaming-A15-FA506QM-FA
deep@deep-ASUS-TUF-Gaming-A15-FA506QM-FA


http://127.0.0.1:3000
Sign in
SmartHome
Devices
Monitor and control your connected devices
Add Device
Devices
Activity Log
3
TOTAL
0
ONLINE
3
OFFLINE
FAN
fan_1
offline
Offline
CAMERA
camera_1
offline
Offline
LIGHT
light_1
offline
Offline
Live
Sign Out


SmartHome
Devices
Monitor and control your connected devices
Add Device
Devices
Activity Log
3
TOTAL
0
ONLINE
3
OFFLINE
FAN
fan_1
offline
Offline
CAMERA
camera_1
offline
Offline
LIGHT
light_1
offline
Offline
Live
Sign Out
Inspector
Console
Debugger
Network
Style Editor
Performance
Memory
Storage
Accessibility
Application
WS
All
HTML
CSS
JS
XHR
Fonts
Images
Media
WS
Other
Disable Cache
No Throttling
Status
Method
Domain
File
Initiator
Type
Transferred
Size
101
GET
127.0.0.1:8080
/
app.js:35 (websocket)
plain
129 B
0 B
Headers
Cookies
Request
Response
Timings
Stack Trace
All
Filter Messages
Data
Time
{"type":"activity_log","entries":[{"timestamp":1774949256,"username":"admin","device_id":"lig...
14:58:12.471
{"type":"get_devices","token":"8c16f8366dc90c031fff84c878641df329c8288caa8d3d4d63298b...
14:58:13.468
{"type":"device_list","devices":[{"device_id":"fan_1","device_type":"fan","state":"offline","last_s...
14:58:13.469
{"type":"get_devices","token":"8c16f8366dc90c031fff84c878641df329c8288caa8d3d4d63298b...
14:58:16.469
{"type":"device_list","devices":[{"device_id":"fan_1","device_type":"fan","state":"offline","last_s...
14:58:16.469
{"type":"get_log","token":"8c16f8366dc90c031fff84c878641df329c8288caa8d3d4d63298bec72...
14:58:17.471
{"type":"activity_log","entries":[{"timestamp":1774949256,"username":"admin","device_id":"lig...
14:58:17.471
{"type":"get_devices","token":"8c16f8366dc90c031fff84c878641df329c8288caa8d3d4d63298b...
14:58:19.490
{"type":"device_list","devices":[{"device_id":"fan_1","device_type":"fan","state":"offline","last_s...
14:58:19.490
{"type":"get_log","token":"8c16f8366dc90c031fff84c878641df329c8288caa8d3d4d63298bec72...
14:58:22.472
{"type":"activity_log","entries":[{"timestamp":1774949256,"username":"admin","device_id":"lig...
14:58:22.472
{"type":"get_devices","token":"8c16f8366dc90c031fff84c878641df329c8288caa8d3d4d63298b...
14:58:22.490


SmartHome
Devices
Monitor and control your connected devices
Add Device
Devices
Activity Log
3
TOTAL
3
ONLINE
0
OFFLINE
FAN
fan_1
on
Toggle State
CAMERA
camera_1
active
Toggle State
LIGHT
light_1
on
Toggle State
Live
Sign Out
Inspector
Console
Debugger
Network
Style Editor
Performance
Memory
Storage
Accessibility
Application
WS
All
HTML
CSS
JS
XHR
Fonts
Images
Media
WS
Other
Disable Cache
No Throttling
Status
Method
Domain
File
Initiator
Type
Transferred
Size
101
GET
127.0.0.1:8080
/
app.js:35 (websocket)
plain
129 B
0 B
Headers
Cookies
Request
Response
Timings
Stack Trace
All
Filter Messages
Data
Time
{"type":"command_ack","success":true,"device_id":"camera_1","action":"active"}
14:58:32.937
{"type":"get_devices","token":"8c16f8366dc90c031fff84c878641df329c8288caa8d3d4d63298b...
14:58:32.938
{"type":"get_log","token":"8c16f8366dc90c031fff84c878641df329c8288caa8d3d4d63298bec72...
14:58:32.938
{"type":"device_list","devices":[{"device_id":"fan_1","device_type":"fan","state":"on","last_seen"...
14:58:32.938
{"type":"activity_log","entries":[{"timestamp":1774949312,"username":"admin","device_id":"ca...
14:58:32.938
{"type":"command","token":"8c16f8366dc90c031fff84c878641df329c8288caa8d3d4d63298bec...
14:58:33.631
{"type":"command_ack","success":true,"device_id":"light_1","action":"on"}
14:58:33.631
{"type":"get_devices","token":"8c16f8366dc90c031fff84c878641df329c8288caa8d3d4d63298b...
14:58:33.632
{"type":"get_log","token":"8c16f8366dc90c031fff84c878641df329c8288caa8d3d4d63298bec72...
14:58:33.632
{"type":"device_list","devices":[{"device_id":"fan_1","device_type":"fan","state":"on","last_seen"...
14:58:33.632
{"type":"activity_log","entries":[{"timestamp":1774949313,"username":"admin","device_id":"lig...
14:58:33.632
{"type":"get_devices","token":"8c16f8366dc90c031fff84c878641df329c8288caa8d3d4d63298b...
14:58:34.499